

Nimbus Nine — SD1 Training Pilot

Workbook 2

Clean, Fix, and Prepare Keys

Difficulty	3 / 10 (gentle)
Estimated time	45 – 60 minutes

Files for this workbook

Start file: 02_workbook_data/workbook2_start.xlsx

Complete file: 02_workbook_data/workbook2_complete.xlsx

Source files required:

- 02_workbook_data/workbook2_start.xlsx

FROM THE FLIGHT DECK

Quartermaster Thessaly Comet says:

“Cleaning data is detective work. Every blank is a question, not a crime.”

YOU ARE DEVELOPING

Skills: S2 (clean data), S3 (organise data), S5 (assess quality)

Knowledge: K10 (data quality), K11 (data cleansing techniques)

Behaviours: B2 (works methodically), B4 (responds positively to feedback)

The mission

MISSION BRIEF

We have three queries imported from Workbook 1.
Now we are going to gently tidy them in Power Query so they are ready to merge.
We will use Power Query the whole way through. No worksheet formulas.
By the end, our keys will be clean enough to join the tables together in Workbook 3.

What you will learn

- How to trim leading and trailing spaces in Power Query.
- How to standardise text casing in Power Query.
- How to replace inconsistent category labels.
- How to investigate blanks before deciding what to do.
- How to remove duplicates safely using Power Query.
- How to check whether a key column is unique.

WHY THIS MATTERS

Clean keys = clean joins.
If our courier_id or route_code values have stray spaces, joins miss rows silently.
Power Query keeps a record of every step, so you can always see what you changed.

Mission 1 — Open the workbook and reopen Power Query safely

1. Open 02_workbook_data/workbook2_start.xlsx
2. Click the Data tab.
3. Click Queries & Connections to open the panel on the right.
4. Find the query Deliveries_Raw.
5. Double-click Deliveries_Raw to open Power Query Editor.

COMMON TRAP

If Power Query Editor is already open from a previous session, just continue from the open window.

If the query is missing, do not panic. Go to Data > Get Data, import 01_source_data/source_primary.xlsx again, and click Transform Data.

PAUSE AND CHECK

Can you see Deliveries_Raw open in Power Query Editor?

On the right, in Query Settings, you should see Applied Steps.

If yes, on we go.

Mission 2 — Trim and clean text

Trim removes leading and trailing spaces. Clean removes hidden non-printing characters.

Trim cargo_type

1. In Power Query Editor, find the cargo_type column header and click on it to select it.
2. Click the Transform tab on the ribbon.
3. Click Format.
4. Click Trim to get rid of extra spaces at the start and end of the field
5. Watch the cargo_type column. Spaces at the start and end should be gone.

Clean cargo_type

6. Make sure cargo_type is still selected.
7. Click Transform, then Format, then Clean to get rid of any messy characters

PAUSE AND CHECK

Look at the Applied Steps panel on the right.

You should now see two new steps: Trimmed Text and Cleaned Text.

Mission 3 — Fix casing

We want each text value to look the same shape every time. Capitalize Each Word turns 'fragile' and 'FRAGILE' into 'Fragile'.

1. Stay on the cargo_type column header.
2. Click Transform, then Format, then Capitalize Each Word.
3. Click the status column header.
4. Click Transform, then Format, then Capitalize Each Word.

Now we need to be careful with one tricky value: "In Transit".

Capitalize Each Word will already keep that one tidy.

COMMON TRAP

Capitalize Each Word changes the live data. If your Applied Steps grow too long, do not worry — that is normal.

Each step is a record of something you did. Click any step to time-travel and see the table at that point.

PAUSE AND CHECK

Scroll the status column. Do all values look like 'Delivered', 'In Transit', 'Returned', 'Cancelled'?

If yes, lovely. If not, click the step that fixed casing and check it ran on the right column.

Mission 4 — Investigate blanks

Some parcels have a blank `parcel_weight_kg` value. We will find them, but we will not delete them yet.

1. Click the View tab on the ribbon.
2. Tick Column quality, Column distribution, and Column profile.
3. A small box above each column will show how many values are valid, error, or empty.
4. Find the `parcel_weight_kg` column. Note the percentage of empty rows
5. Click the small filter arrow on the `parcel_weight_kg` column.
6. Tick only (null) to focus on the blank rows.
7. Look at which `delivery_ids` are missing weights.
8. Click the filter arrow again and click Clear Filter to bring everything back.

TRY THIS

Tiny challenge:

Write down two `delivery_ids` with blank weights.

Are they all on the same route? Or all the same status?

Patterns are gold dust.

We are going to keep these blanks for now. Captain Vega will decide later. Removing data without asking is rarely the right move.

Mission 5 — Remove duplicates carefully

In this mission, you are going to remove duplicate delivery records.

The column we are checking is **delivery_id**.

This is important because **delivery_id should be unique**.

That means each delivery should only appear **once** in the data.

At the moment, there are two duplicate pairs:

- row 24 and row 25
- row 43 and row 44

After removing the duplicates, the table should reduce from **52 rows to 50 rows**.

1. Click the `delivery_id` column header to select it. This tells Power Query: “Only check this column when looking for duplicates.”
2. Click the Home tab.
3. Click Remove Rows.
4. Click Remove Duplicates.
5. Watch the row count at the bottom of the editor — it should drop from 52 to 50.

COMMON TRAP

Always select the column you want to dedupe BEFORE clicking Remove Duplicates. If you remove duplicates without selecting a column, Power Query will only remove rows that are exact across every column. If your numbers look wrong, click Undo and try again. Mistakes are normal.

PAUSE AND CHECK

Row count at the bottom of the editor should now read 50. If yes, well done — your `delivery_id` is now your tidy unique key.

Mission 6 — Remove duplicates carefully

In this mission, you are not going to remove any rows.

There are **no full-row duplicates** in this table.

However, it is important to know what you would do **if** you had a larger table and needed to remove rows that were completely identical.

A full-row duplicate means the **whole row is exactly the same** as another row.

That means every value matches, including:

- delivery_id
 - delivery_date
 - customer details
 - delivery status
 - delivery cost
 - every other column in the table

If you did need to remove full-row duplicates, you would do this:

1. Click the small **table icon** in the top-left corner of the Power Query preview.
This selects the **whole table**, not just one column.
This tells Power Query:
“Check every column before deciding whether a row is a duplicate.”
2. Click the **Home** tab.
3. Click **Remove Rows**.
4. Click **Remove Duplicates**.
5. Watch the row count at the bottom of the editor.

The row count should only reduce if Power Query finds rows where **every value in the row is identical**.

COMMON TRAP

Do not select only **delivery_id** if your aim is to remove full-row duplicates.

If you select only **delivery_id**, Power Query will remove rows based only on repeated delivery IDs.

That is useful in some tasks, but it is not the same as removing full-row duplicates.

For full-row duplicates, Power Query must check the **whole table**.

Mission 7 — Check and prepare keys

A key is the column we will use to match rows between tables. Before merging, we want to be sure each key behaves itself.

Rename and finalise Deliveries

1. In Query Settings, change the query name from Deliveries_Raw to Deliveries_Clean.
2. Click Home.

Clean the Couriers query

3. In Queries, click Couriers_Raw.
4. Click courier_name. Click Transform > Format > Trim and Transform > Format > Clean.
5. Click ship_class. Click Transform > Format > Trim and Transform > Format > Clean.
6. With ship_class still selected, click Transform > Format > Capitalize Each Word.
7. In Query Settings, rename the query to Couriers_Clean.

Clean the Routes query (light touch)

13. In Queries click Routes_Raw.
14. Look at each column. Routes is a small reference table and is usually tidy.
15. If you spot anything odd, apply Transform > Format > Trim.
16. In Query Settings, rename the query to Routes_Clean.
17. Click Home, then Close & Load

PAUSE AND CHECK

You should now have three cleaned queries: Deliveries_Clean, Couriers_Clean, Routes_Clean although the spreadsheet names may still reference the old Raws – this is fine

Note your changes

Click the Cleaning_Notes sheet. Jot down what you changed in each query and why. This is the kind of note any good data technician keeps. Future-you will thank you.

What you can now do

- Trim and clean text columns in Power Query.
- Standardise text casing using Capitalize Each Word.
- Replace inconsistent values with Replace Values when needed.
- Investigate blanks calmly using Column Quality.
- Remove duplicates by selecting the right key first.
- Rename queries clearly and load them with the right load option.

NICE WORK

That was Workbook 2.

Your keys are tidy. Your text is consistent. Your duplicates are gone.

If you want to compare against the worked version, open
02_workbook_data/workbook2_complete.xlsx.

Tea break. You earned it.